Juhyun Bae

M.S. Candidate, Electrical Engineering
Deep Image Processing Laboratory, Korea University
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Research Profile

Research interests in image restoration, computational photography, and learning-based data augmentation.

Education

Korea University **Seoul, Korea**
M.S. Candidate in Electrical Engineering *Mar. 2025 – Feb. 2027 (Expected)*

- Advisor: Prof. Seung-Won Jung
- Deep Image Processing Laboratory

Seoul National University of Science and Technology (SeoulTech) **Seoul, Korea**
B.S. in Computer Science and Engineering *Mar. 2020 – Feb. 2025*

- GPA: 4.38 / 4.50
- Graduated as Valedictorian (Ranked 1st in the Department)

Research Experience

Deep Image Processing Laboratory, Korea University **Seoul, Korea**
Graduate Researcher *Mar. 2025 – Present*

- Conduct research on image restoration, computational photography, and learning-based data augmentation.
- Developed a spatially adaptive deblurring framework for display mura compensation in collaboration with LG Display.
- Developing a trajectory-aware differentiable blur augmentation framework for robust image deblurring.

Computer Vision Laboratory, SeoulTech **Seoul, Korea**
Undergraduate Researcher *Feb. 2024 – Nov. 2024*

- Conducted research on image aesthetics assessment and captioning using vision-language models.
- Developed and optimized machine learning force field models for the Samsung AI Challenge 2024, achieving 2nd place.

Center for Artificial Intelligence, KIST **Seoul, Korea**
Undergraduate Research Intern *Feb. 2023 – Sep. 2023*

- Developed YOLOv7-based PPE detection and second-stage classification models for laboratory safety inspection.
- Optimized the real-time inference pipeline with TensorRT for deployment in an access-control kiosk.

Publications

International Journals

1. Geon-Ho Park[†], **Juhyun Bae**[†], Jinah Kim, Changeui Hong, and Seung-Won Jung, “DnSAD: Defocus and Spatially Adaptive Deblurring for Mura Compensation in Display Panels.”
   Journal of the Society for Information Display, 34(3):99–108, 2026. [†] *Co-first authors*

2. Jinah Kim, Geon-Ho Park, **Juhyun Bae**, and Seung-Won Jung, “Intrinsic Decomposition-Based Curriculum Learning for Low-Light Image Enhancement.”
   IEEE Signal Processing Letters, 32:4019–4023, 2025.

Domestic Conference Papers

1. **Juhyun Bae**, et al., “Layer-wise Embedding Analysis of Multimodal Language Model Decoders for Visual Information Processing.”
   Institute of Electronics and Information Engineers (IEIE), 2025.

2. **Juhyun Bae**, Y. Sung, Y. Yuk, and H. Jang, “Chatbot for Diagnosis of Pet Diseases.”
   Korea Information Processing Society (KIPS), 2022.

Selected Research

Current Research
Trajectory-aware Differentiable Blur Augmentation for Image Deblurring **2026 – Present**

- Developed a trajectory-aware reblur network with physically interpretable camera and object motion modeling.
- Developed learnable differentiable blur augmentation jointly optimized with the deblurring objective.
- Manuscript in preparation for ICLR 2027.

Published Research
DnSAD: Defocus and Spatially Adaptive Deblurring for Display Mura Compensation **2025 – 2026**

- Developed a PSF-aware deblurring framework for spatially varying display mura compensation.
- Integrated measured PSF maps with deep Wiener deconvolution and spatial attention.
- Co-first author, *Journal of the Society for Information Display (JSID)*, 2026.

Previous Research
Image Aesthetics Assessment using Vision-Language Models **2024**
SeoulTech

- Conducted research on vision-language models for image aesthetics assessment and captioning.

Machine Learning Force Fields **2024**
Samsung AI Challenge

- Developed and optimized MACE-based graph neural networks for molecular energy and force prediction.
- Developed a Gaussian Process Regression-based framework for out-of-distribution uncertainty estimation.

Vision-based Safety Equipment Inspection **2023**
KIST

- Developed a YOLOv7-based multi-stage PPE inspection system with TensorRT deployment for real-time laboratory access control.

Awards & Honors

2nd Place, Samsung AI Challenge – Machine Learning Force Fields 2024
Honorable Mention, Hanium ICT Mentoring Project Contest 2022
1st Place, SeoulTech DDR Camp 2022

Scholarships

- Academic Excellence Scholarship, SeoulTech – Full tuition (Spring 2021)
- Academic Excellence Scholarship, SeoulTech – Half tuition (Fall 2021, Fall 2022, Spring 2023)

Technical Skills

Programming Python, C++, MATLAB
Deep Learning PyTorch, CUDA, Distributed Training, Mixed Precision
Deployment TensorRT, Docker, Linux, Git
Languages Korean (Native), English (TOEIC 970, 2023)